Machine Learning II

Data Science M.Sc.

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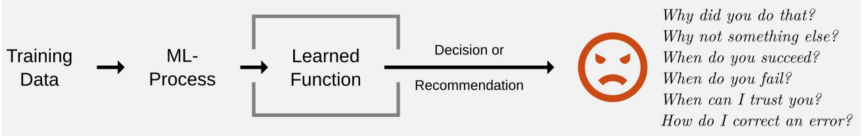
An Introduction to **Interpretable Machine Learning**

Deconstructing the *Black Box* with Explainable AI[1]

[1] We will use both the terms *Explainable AI (XAI)* and *Interpretable ML (IML)* synonymously.

Why Interpretability matters

Often a ML process produces a learned function that is a black box. This leads to critical *unanswered* questions from the user.



Critical scenarios: a ML model …

- denies a loan.
- flags a tumor.
- recommends a prison sentence[2].
- … .

[2] Ryberg, J. (2025). *Artificial intelligence at sentencing: when do algorithms perform well enough to replace humans?*. AI and Ethics, 5(2), 1009-1018. https://link.springer.com/content/pdf/10.1007/s43681-024-00442-5.pdf

Why Interpretability matters

Identification of incorrect mechanisms

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Classified as horse

Not classified as horse

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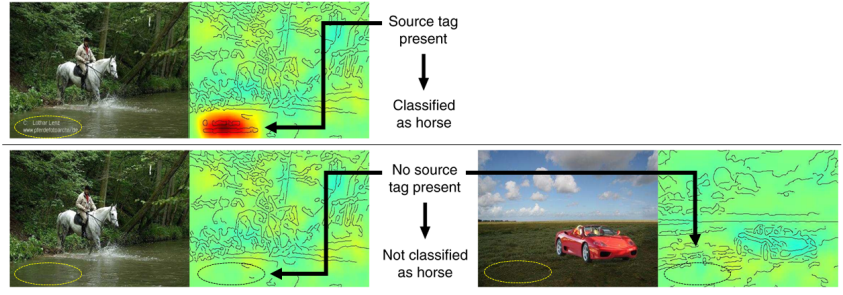
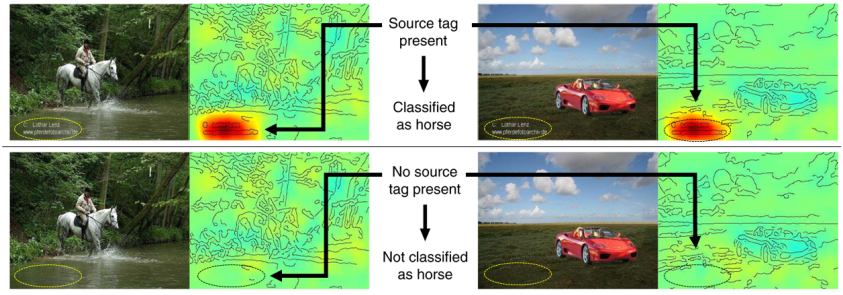
Identification of incorrect mechanisms



Why Interpretability matters

Identification of incorrect mechanisms



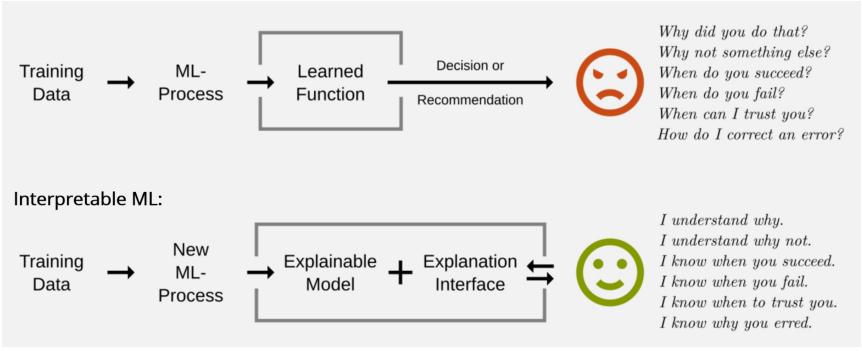
Sebastian Lapuschkin et al. *Unmasking clever hans predictors and assessing what machines really learn.* Nature communications, 10(1), 2019.
https://www.nature.com/articles/s41467-019-08987-4

The Goals of Interpretability

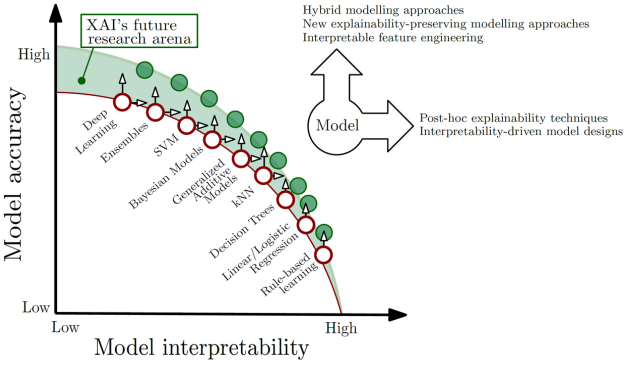
1. **Improving the Model**
   - **Debugging:** Find "Clever Hans" predictors, i.e. models that "cheat" using shortcuts.
   - **Feature Engineering:** Use feature insights to get new ideas for creating better features.
   - **Sanity Checks:** Verify that feature effects match domain knowledge (e.g., "Does 'Age' impact risk in a way that makes sense?").
2. **Justifying the Model and its Predictions**
   - **Build Trust:** Justify predictions to various stakeholders (managers, clients, regulators).
   - **Enable Recourse:** Allow a "decision subject" (e.g., a loan applicant) to understand *why* a decision was made and how they might contest it.
   - **Regulatory Compliance:** Meet legal and regulatory requirements (like GDPR's "Right to Explanation") by proving a model is safe and fair.
3. **Discovering New Insights**
   - **Knowledge Generation:** Use the model to learn new things about the world.
   - **Inform Decisions:** Move beyond just *predicting* churn to *understanding* what drives it, e.g. allowing a marketing team to take targeted action.

Source: Based on "Goals of Interpretability" from Christoph Molnar: `https://christophm.github.io/interpretable-ml-book/goals.html`

From Predictive ML to Interpretable ML

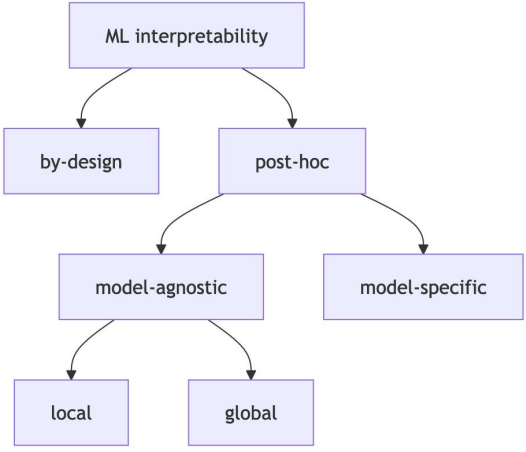


The Performance-Interpretability Trade-off



Source: Arrieta et al: *Explainable Artificial Intelligence (XAI): Concepts, Taxonomies, Opportunities and Challenges toward Responsible AI.* Information Fusion. 2020. (`https://doi.org/10.1016/j.inffus.2019.12.012`)

A Taxonomy of Interpretability Methods



Source: "Methods overview" from Christoph Molnar: `https://christophm.github.io/interpretable-ml-book/overview.html`

Interpretable models

Interpretability by Design (Intrinsic): White box models

- We choose a ML algorithm that is based on a representation *understandable* by humans.
- The model itself is the explanation, i.e. model and/or its parameters can be interpreted.

Examples:

- Linear / Logistic Regression: A one-unit increase in x_j changes the outcome y by β_j, holding all else constant.
- GAMs: The visualized splines show the relationships between x and y
- Decision Trees: A set of explicit `IF-THEN-ELSE` rules partitioning the feature space.
- Decision Rules: The rules themselves.

Post-hoc Interpretability

We first train a complex, “black box” model (like Random Forests, XGBoost or Neural Network) and then apply a separate method *afterwards* to explain it. We distinguish:

Model-Specific post-hoc methods

- Can only be used for a specific model class.
- Example: Gini importance in a Random Forest.

Model-Agnostic post-hoc methods

- Can be used on *any* machine learning model, regardless of its internal structure.
- This is the most flexible and powerful category.
- Examples: PDP, SHAP, Permutation Importance.

Scopes of Interpretability

Global Methods:

- Explain the *entire model's* behaviour.
- Questions:
  - **Feature importance:**
    Which features are carrying the most important informations?
    Which features contribute the most to model quality?
  - **Feature effects:**
    What is the relationship between feature X and outcome Y?
    How does X affect Y?

Local Methods:

- Explain a *single, individual prediction*.
- Question: Why was *this specific customer* denied a loan?

Model-Agnostic XAI

In the following, we focus on **post-hoc model-agnostic**[3] methods.

Advantages:

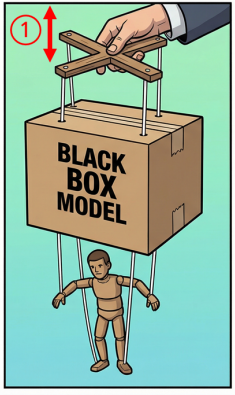
- **Flexibility:** Use any ML model you like (e.g., XGBoost, Neural Net).
- **Standardization:** The interpretation method is the same.
- **Comparability:** Different models can be evaluated by the same XAI method.

General Idea:

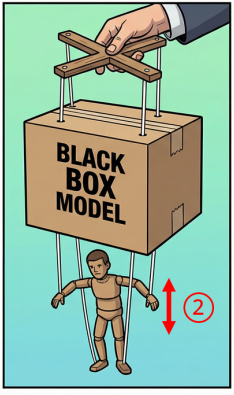
- We only need the **Prediction Function** $\hat{y} = f(\mathbf{X})$ of our trained (black box) model. We don't need to understand and know internal model parameters.
- We gain understanding by investigating how a change in input feature values changes the model's outcome, i.e. its predictions.

[3] In the very end of this chapter we will come back to one model-specific method.

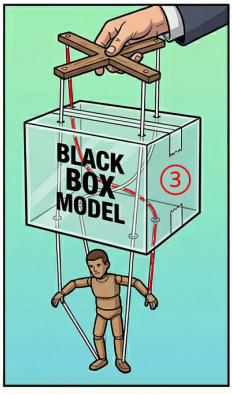
Model-Agnostic XAI



Vary input features



Observe how predictions change



Understand model mechanism

Overview of Model-Agnostic XAI Methods

Method	Global Feature Importance	Global Feature Effects	Local Explanation
Permutation Feature Importance (PFI)	**Yes** (Provides a ranked list of what's important)	**No** (Tells you *if* a feature matters, not *how*)	**No**
Partial Dependence Plot (PDP)	**Indirectly** (not recommended)	**Yes** (Shows the average "what-if" effect. **Note: Unreliable with correlated features**)	**No**
Accumulated Local Effects (ALE)	**Indirectly** (not recommended)	**Yes** (Shows the average effect. **Designed to be robust to correlated features**)	**No**
SHAP	**Yes** (By aggregating local values)	**Yes** (Using summary and dependence plots)	**Yes** (This is its foundation)

Remarks:

- The table shows why SHAP is so popular. It is a unified framework that provides a clear solution for all three goals: it builds local explanations which can then be aggregated to create robust global feature importance and global feature effects.
- ALE is the modern, preferred alternative to PDP for only analyzing global feature effects, as it correctly handles correlated features.

Permutation Feature Importance (PFI)

Core Idea: How much does the model's performance get worse when we *break* a feature's link to the target by randomly shuffling it?

Algorithm:

1. Train ML model and calculate its baseline test error e_{orig} using an appropriate loss function L.
2. For *selected* feature S create a *broken* dataset $\mathbf{X}_{S,\text{perm}} = (S, \mathbf{C})$ by **randomly shuffling** values in feature S and leaving all other *complementary* features $\mathbf{C}$ unchanged.
3. Calculate the new error on this permuted data: $e_{S,\text{perm}} = L(y, f(\mathbf{X}_{S,\text{perm}}))$.
4. The importance of feature S is the **difference** (or ratio) between the errors:

$$FI_S = \Delta e_S = e_{S,\text{perm}} - e_{\text{orig}}$$

5. Do so for every feature and compare the FI values.

Remarks:

- **Pros:** Model-agnostic, very intuitive, provides a single global importance score.
- **Cons: Problematic with correlated features**.

Partial Dependence Plot (PDP)

Core Idea: What is the *average* model prediction as one feature changes?

Algorithm:

1. Choose a feature S and a grid of values for it (usually covering its range).
2. To get the PDP value for a specific grid value of $S = s^*$:
   - Take *all n observations* from your dataset.
   - **Force** the column S to the specific grid value s^* for *every single observation*, leaving all other features $\mathbf{C}$ as-is.
   - Calculate the **average** of all n predictions.
3. Plot these averages for each grid value.
4. The PDP value for a feature value s^* is:

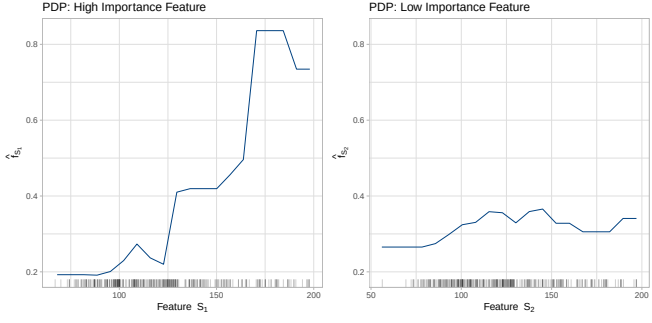
$$\hat{f}_S(s^*) = \frac{1}{n} \sum_{i=1}^{n} f(S = s^*, \mathbf{C}_i)$$

where $\mathbf{C}_i$ are the values of all *other* features $\mathbf{C}$ for the i-th observation.

Deriving Feature Importance from PDP

Core Idea: A feature is "important" if changing its value causes a *large change* in the model's average prediction.

- A feature with a **flat PDP line** is **unimportant** because changing its value doesn't affect the model's output.
- A feature with a **steep or "wiggly" PDP line** is **important**.
- We quantify "wiggliness" by measuring the **dispersion** of the PDP values.



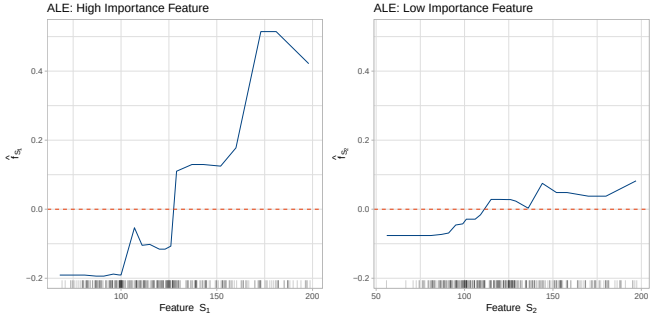
Accumulated Local Effects (ALE)

Core Idea: A robust alternative to PDP that **handles correlated features** by averaging *changes* in predictions, not the predictions themselves.

Algorithm:

1. Select a feature S and divide its range into k small intervals $I_k = (s_k^{low}, s_k^{upp}]$.
2. Use *only* the data points i with $s_i \in I_k$:
   i) By replacing s_i with the interval's boundaries, Calculate the i **differences in predictions**
   $$\Delta f(s_i) = f(s_k^{upp}, \mathbf{C}_i) - f(s_k^{low}, \mathbf{C}_i)$$
   ii) For an interval k, $\Delta f(s_k)$ is the average of all $\Delta f(s_i)$:
   $$\Delta f(s_k) = \frac{1}{N_i} \sum_i \Delta f(s_i)$$
   iii) Do so for all intervals.
3. **Accumulate** these average differences $\Delta f(s_k)$ across all intervals to build the ALE plot:
   $$f(s) = \sum_k \Delta f(s_k)$$
4. Center $f(s)$, so that the mean effect is zero.

Accumulated Local Effects (cont)



Remarks:

- **Pros: Robust to correlated features**, fast
- **Cons:** Slightly more complex to explain than PDP, ALE can be interpreted as the main effect of the feature at a certain value compared to the average prediction of the data.
- Feature Importance: Can be inferred from the plot's effect size as with PDP

- Lundberg et al: *A unified approach to interpreting model predictions.* Advances in neural information processing systems. 2017.
  `https://proceedings.neurips.cc/paper/2017/file/8a20a8621978632d76c43dfd28b67767-Paper.pdf`
- `https://github.com/slundberg/shap`

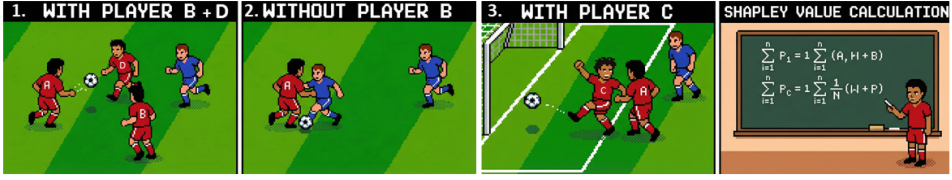
Why SHAP?

SHAP (SHapley Additive exPlanations) has become an industry standard.

- It is a framework to explain **individual predictions** (local).
- Its global interpretation methods are built by aggregating these individual effects.
- This provides a **common ground for local and global explanations**.
- It has a **solid theoretical foundation** based on Shapley values from coalitional game theory.
- SHAP is a framework to estimate Shapley values in the XAI context and marks a change in popularity and usage of Shapley values for XAI.
- The SHAP concept is **model-agnostic**.

Shapley values and coalitional game theory

Consider a soccer match and a coalition of players A, B, C and D:



1. Players A, B and D are working together to advance the ball down the soccer field.
2. Player A is without team-mates continuing the attack on its own.
3. Player A is joined by Player C. A passes to C and C scores.

Question: What is the contribution of player A achieving the goal?

Coalitional game theory

- **The “Game”:** A cooperative effort where multiple “players”, i.e. the coalition, work together to achieve a single outcome or “payout”.
- **The “Payout”:** The total value generated by the coalition.
- **The “Players”:** The specific participants in a specific game.
- **The Goal:** To distribute the total payout fairly among the players based on their individual contributions.

- **The Solution (Shapley Value):**
  It calculates the **average marginal contribution** of a player across all possible coalitions.

Mapping Game Theory to Machine Learning

Game Theory Concept	Machine Learning Equivalent
The Game	The model making a prediction for a single instance.
The Players	The **feature values** of that instance (e.g., `Age`=30, `Income`=50k).
The Payout	The difference between the **actual prediction** and the **average prediction** ($f(x) - E[f(x)]$).
Coalition	A subset of features (some feature values are "present", others are "absent" or marginalized).
Marginal Contribution	The change in prediction when a specific feature is added to a coalition.
Shapley Value	The feature's contribution to the prediction (its "importance" for this specific instance).

Shapley values: a ML example

Example: **apartment price** predictions[4]

Features:

- `park_near_by` (yes/no)
- `flat_size` (square meters)
- `floor` (1st, 2nd, 3rd, ...)
- `cats_allowed` (yes/no)

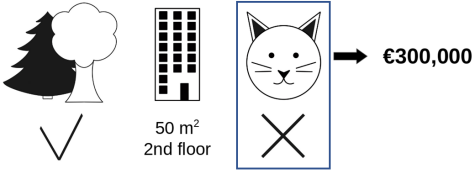
We consider a specific flat (with park near by, 50 m^2, 2nd floor and no cats allowed).

Questions:
To what extent does the fact that cats are not allowed contribute to the predicted price of 300.000 EUR?

[4] source: `https://christophm.github.io/interpretable-ml-book/shapley.html`

Shapley values: a ML example

Example: **apartment price** predictions[5]



What is the contribution of `no-cats-allowed` to the predicted price?

[5] source: `https://christophm.github.io/interpretable-ml-book/shapley.html`

Shapley values: a ML example

Definition: The Shapley value is the average marginal contribution of a feature value across **all possible coalitions** of features.

# Features	# Coalitions	Sets of coalitions **S** without feature **X**:
0	1	
1	3	50m² | 2nd floor
2	3	50m² | 2nd floor | 50m² 2nd floor
3	1	50m² 2nd floor

Shapley values: a ML example

Definition: The Shapley value is the **average marginal contribution** of a feature value across all possible coalitions of features.

# Features	# Coalitions	Sets of coalitions **S** without feature **X**:	Marginal contribution of **X**	Weighted Average
0	1		 $= \Delta_{\lvert S\rvert=0}$	$\Delta_{\lvert S\rvert=0}$
1	3		 $= \Delta_{\lvert S\rvert=1}$	$\Delta_{\lvert S\rvert=1}$
2	3		 $= \Delta_{\lvert S\rvert=2}$	$\Delta_{\lvert S\rvert=2}$
3	1		 $= \Delta_{\lvert S\rvert=3}$	$\Delta_{\lvert S\rvert=3}$

Φ

Shapley values for ML models

The vector of Shapley values $\Phi(X_j)$ of a feature X_j are its contribution to the predicted payout, weighted and summed over all possible feature value combinations:

$$\Phi(X_j) = \sum_{S \subseteq N \setminus X_j} \underbrace{\frac{|S|!\,(|P| - |S| - 1)!}{|P|!}}_{\text{weighting}} \underbrace{[f(S \cup X_j) - f(S)]}_{=\Delta_S(X_j)}$$

with

$f(S \cup X_j)$ ... outcome of ML model for set of features S with consideration of feature X_j

$f(S)$ ... outcome of ML model for set of features S without consideration of feature X_j

P ... set of all features

$|P|$ ... number of all features

S ... subset of features in specific coalition

$|S|$ ... number of features in coalition S

X_j ... feature of interest

Estimation of Shapley values

- All possible coalitions S of feature values have to be evaluated with and without the j-th feature to calculate the exact Shapley value.
- For more than a few features, the exact solution to this problem becomes problematic as the number of possible coalitions exponentially increases as more features are added.
- It requires retraining the model on $2^{|P|}$ feature coalitions.
- **SHAP** provides efficient way(s) to estimate Shapley values.

Estimation of Shapley values

There are three ways to estimate Shapley values based on SHAP for tabular data:

- KernelSHAP: The original estimation method based on a theoretical connection to LIME[6]. Since it is slow, it isn't used any more in most software packages.
- Permutation Method:[7] Most efficient **model-agnostic** estimation method. The idea is to sample cleverly from all possible coalitions by only considering a small number of sets with a permuted feature sequence. Default model-agnostic software implementation (e.g. `DALEX` package in R).
- TreeSHAP[8]: TreeSHAP was introduced as a fast, **model-specific** alternative to KernelSHAP. Integrated directly into boosting frameworks like XGBoost, Catboost and LightGBM.

We will briefly discuss TreeSHAP, because
Gradient Boosted Trees + TreeSHAP = "Industry Standard".

[6] = Local Interpretable Model-agnostic Explanations. Not covered in this lecture.

[7] Mitchell, Rory, Joshua Cooper, Eibe Frank, and Geoffrey Holmes. 2022. "Sampling Permutations for Shapley Value Estimation." Journal of Machine Learning Research 23 (43): 1–46. `http://jmlr.org/papers/v23/21-0439.html`.

[8] Lundberg, Scott M., Gabriel G. Erion, and Su-In Lee. 2019. "Consistent Individualized Feature Attribution for Tree Ensembles." arXiv. `https://doi.org/10.48550/arXiv.1802.03888`.

TreeSHAP

- This is a **fast, model-specific** alternative for tree-based models (XGBoost, Random Forest) leveraging the tree structure for efficient computation.
- It computes **exact Shapley values** and reduces the computational complexity from $O(TL2^{|P|})$ to $O(TLD^2)$, where T is the number of trees, L is the maximum number of leaves in any tree, and D is the maximal depth of any tree.[9]
- Since the number of considered features in a single tree is limited by the number of knots, a majority of features is absent in a specific tree. Therefore only a small set of coalitions S alter the predictions, i.e. influence the Shapley values.
- TreeSHAP only focuses on this small set of relevant coalitions.

[9] Especially for Gradient Boosted Trees we use weak learners, i.e. tree stumps, with very small D.

SHAP results

Outcome: Averaged contribution Φ_{ij} to $\hat{y}_i = f(\mathbf{X}_i)$ for every observation i and feature j:

	feature 1	feature 2		feature p
obs 1	$\Phi_{1,1}$	$\Phi_{1,2}$	$\dots$	$\Phi_{1,p}$
obs 2	$\Phi_{2,1}$	$\Phi_{2,2}$	$\dots$	$\Phi_{2,p}$
	$\vdots$	$\vdots$	$\ddots$	
obs n	$\Phi_{n,1}$	$\Phi_{n,2}$		$\Phi_{n,p}$

Learnings:

- row values = **microscopic explanations**: how does each feature $x_{i,\cdot}$ contribute to a specific prediction $\hat{y}_i$
- column values = **feature effects**: how do the contributions $\Phi_{\cdot,j}$ depend on feature values $x_{\cdot,j}$
- sum of absolute column values = **feature importance**: inferred from feature effect size

SHAP: example

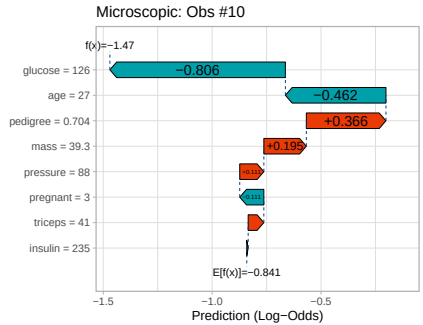
- XGBoost model for the prediction of a binary outcome (`diabetes: yes/no`).
  See `?mlbench::PimaIndiansDiabetes2` for data documentation and exercise sheet covering XGBoost.
- Data set with 768 observations and 8 features.
- Shapley values can be explicitly obtained via

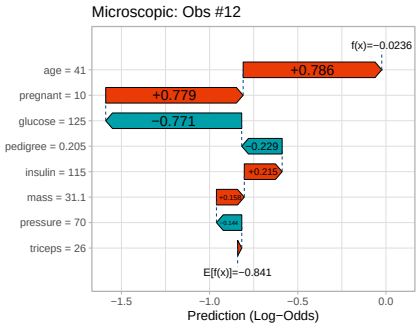
```
xgboost:::predict.xgb.Booster(..., predcontrib = TRUE)
```

	pregnant	glucose	pressure	triceps	insulin	mass	pedigree	age	BIAS
[1,]	-0.046	-1.068	-0.080	-0.196	0.073	-0.190	-0.462	-1.032	-0.841
[2,]	0.146	0.847	0.260	0.109	0.307	0.278	0.643	0.360	-0.841
[3,]	-0.007	-0.942	0.502	0.153	0.518	0.203	-0.065	-0.084	-0.841
[4,]	-0.079	1.910	-0.088	0.068	0.267	0.090	-0.043	0.578	-0.841
[5,]	-0.058	2.598	-0.014	-0.013	0.047	0.077	-0.107	0.411	-0.841
[6,]	-0.178	2.965	-0.086	0.000	0.327	-0.679	0.076	0.356	-0.841
[7,]	0.075	-0.283	0.136	0.037	0.209	0.892	0.135	0.453	-0.841
[8,]	-0.141	-1.074	0.263	-0.017	-0.889	0.136	-0.491	0.142	-0.841
[9,]	0.041	-0.469	-0.074	0.017	0.395	0.287	0.465	0.546	-0.841
[10,]	-0.111	-0.806	0.111	0.072	0.006	0.195	0.366	-0.462	-0.841
[11,]	0.581	0.146	0.169	0.079	0.441	0.339	-0.198	0.877	-0.841
[12,]	0.779	-0.771	-0.144	0.023	0.215	0.158	-0.229	0.786	-0.841

- Last column `BIAS` contains the overall mean $E[f(x)] = \bar{y}$. The other columns showing the Shapley values are related to this mean.

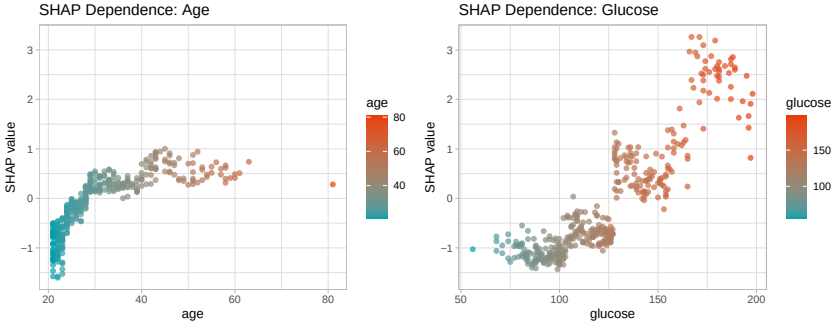
Microscopic explanations: waterfall plots





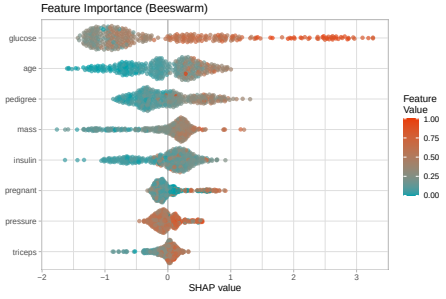
- The prediction $\hat{f}(x_{10}) = -1.47$ and $\hat{f}(x_{12}) = -0.024$.
- The difference between $\hat{f}(x_i)$ and $E[f(x)]$ is the sum of the individual Shapley values.
- The observed feature values x_{ij} are given as numeric values on the plot's y-axis.
- Plots are obtained using the `shapviz` package with which you will work in the exercise session.

Feature Dependence



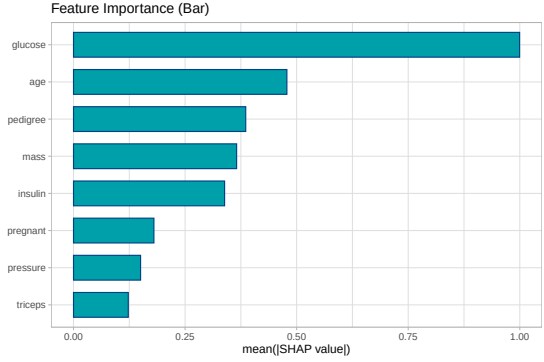
- SHAP values on the y-axis plotted against observed feature values on the x-axis.
- The dependence between feature X_j and the predicted outcome is visualized.
- The larger the effect size, i.e. the spread of SHAP values on the y-axis, the more the feature contributes to the variation in $\hat{y}_i$.

Feature Importance: Beeswarm Plot



- The features are listed from top to bottom based on their overall importance.
- The color represents the actual feature value (Blue = Low, Red = High). By looking at where the Red dots land, you can see the direction of the relationship: if Red dots are on the right, higher feature values increase the predicted outcome.
- The horizontal spread of the dots shows the variance of the effect. A wide spread indicates that the feature acts differently for different patients, while a tight cluster suggests an almost consistent effect for all observations.

Feature Importance: Bar Plot



- Feature importance is obtained as mean of absolute Shapley values for each feature.
- The larger the effect size, the larger the feature importance.

SHAP

Consistent concept for *local* and *global* explanations:

Summary

- **Interpretability is key** for model acceptance and improvement.
- There is a **trade-off** between predictive power and interpretability.
- **Post-hoc techniques** (like PFI, PDP, ALE, LIME, and SHAP) shed light on black box algorithms.
- The **SHAP framework** is a powerful, unified approach providing Local & global explanations.
- The combination of **Gradient Boosted Trees** and **TreeSHAP** is SOTA.

IML vs. Statistical Inference

- IML methods are **descriptive tools**. They describe the inner workings of the trained machine learning model, not the statistical properties of the population from which the data was drawn.
- Unlike classical statistical models, post-hoc IML results do not provide **p-values**, **standard errors**, or **hypothesis tests**.
- One can only assume that a feature will be very likely *"statistically significant"* simply if it has a high importance score or shows a non-flat PDP curve.